Cubic B-Spline Outline of the Ferrari logo

Name: Chris Varghese Mathew | **Roll No:** B240029ME | **Class:** ME-02

1 Objective

The aim of this assignment was to use a cubic B-spline (order $k = 4$) to trace the outline of the silhouette of the Ferrari brand logo. The provided Python notebook (`ferrari_logo.ipynb`) already contained a working B-spline evaluator. The task was to replace the control points to obtain the required shape.

The shape chosen is an original silhouette of the Ferrari brand logo, the prancing horse.

2 Background: B-spline curves

2.1 Definition

A B-spline curve of order k (degree $k-1$) with n control points $P_0 \dots P_{n-1}$ is

$$C(t) \;=\; \sum_i P_i \, N_{i,k}(t)$$

where $N_{i,k}(t)$ are the B-spline basis functions defined on a knot vector $t_0 \le t_1 \le \cdots \le t_{n+k-1}$. The basis functions come from the Cox–de Boor recursion:

$$N_{i,1}(t) = \begin{cases} 1 & t_i \le t < t_{i+1} \\ 0 & \text{otherwise} \end{cases}$$

$$N_{i,k}(t) = \frac{t - t_i}{t_{i+k-1} - t_i} \, N_{i,k-1}(t) \;+\; \frac{t_{i+k} - t}{t_{i+k} - t_{i+1}} \, N_{i+1,k-1}(t)$$

Terms with a zero denominator are taken as zero.

2.2 Properties used in this work

- **Local support.** Each control point only influences $k = 4$ knot spans. Moving one point changes the curve locally, which makes manual shape design practical.
- **Convex hull property.** Every point on a span lies inside the convex hull of the k control points active on that span.
- **Smoothing.** The curve does not pass through its control points; it is pulled toward them. Sharp features in the control polygon are therefore rounded off.
- **Continuity.** For a cubic B-spline with distinct knots the curve is C^2 continuous (position, tangent and curvature are all continuous).

- **Open-uniform knot vector.** The first and last knots are repeated k times and the interior knots are equally spaced. This forces the curve to start at the first control point and end at the last one.

3 The provided code

The notebook is organised as follows.

1. `basis_function(i, k, t, knots)` evaluates $N_{i,k}(t)$ with the Cox–de Boor recursion. It includes a special case so the curve is defined at the final knot $t = 1$.
2. `generate_open_uniform_knots(n_ctrl, k)` builds a knot vector of length $n+k$. The first k knots are 0, the last k knots are 1, and the $n-k$ interior knots are evenly spaced.
3. `b_spline_curve(control_points, k, num_samples, knots)` samples the parameter t uniformly on $[0, 1]$, sums $P_i \cdot N_{i,k}(t)$ over all control points at each sample, and returns the sampled curve and the knot vector used.
4. The input section holds the `control_points` list. A flag, `RUN_SHARP_DEMO`, selects either the L-shaped corner demonstration (which uses a custom knot vector with an interior knot repeated three times) or the main shape.
5. The plotting section draws the control polygon (grey), the B-spline (blue) and the start/end point, then saves the figure as `b_spline_curve.jpg` and prints the control points and knot vector so they can be copied into a report.

The B-spline code itself was not changed. The edits made to the notebook were to the main-shape `control_points` list, the plot title, the figure size (made taller for the upright horse), and `num_samples` (set to 1200 so the sharp tips are drawn smoothly).

4 Design method

4.1 Coordinate system and ordering

The figure was designed on a grid of roughly 0 to 150 in x and 0 to 165 in y, with y pointing up so the plot is not upside down. The ground is $y = 0$. The control points are listed in a single closed loop, clockwise when viewed on screen, starting at the front-bottom corner (the toe) of the forward hind hoof at $(72, 0)$ and finishing at the same point so the outline closes. Sharp corners (hoof edges, ear tips, mane tufts, tail tips, the elbow and the crotch) are produced by repeating a control point three times; the full outline uses 206 control points including these repeats.

The path visits the following regions in order:

1. Front edge of the forward hind leg: from the hoof toe up through the pastern, hock and stifle.
2. Belly: from the flank up and to the left, to the elbow.
3. The first (lower, folded) foreleg: underside of the forearm, knee, cannon and hoof, then back up its front edge and along the top of the forearm.
4. The second (upper, raised) foreleg: underside of the forearm, knee, cannon and hoof, then back up its front edge, over the knee and along the top of the forearm to the shoulder.

5. Chest and throat, then the jaw, chin, muzzle, nose bridge and forehead.
6. Both ears and the poll.
7. Mane: four spiked tufts along the back of the neck.
8. Back line, sloping down to the croup and the tail root.
9. Tail: out along its upper edge to a pointed tip, then back and forth through two further pointed locks to the buttock.
10. The rear (bent) hind leg: back edge down through the hock to the hoof, the hoof bottom, then up its front edge to the crotch.
11. The forward hind leg, back side: from the crotch down to the heel, then the ground line back to the start point.

Walking around the outline once, without crossing, is what lets a single open B-spline describe a closed silhouette.

4.2 Why a plain polygon is not enough

Because a B-spline is pulled toward but not through its control points, tracing a shape by placing points exactly on the desired outline produces a curve that cuts inside every bend. Two techniques were used to compensate:

- **Overshoot:** Points at convex bends (the front of the knees, the top of the head, the tail curl) were placed slightly outside the intended outline so the curve lands where it should.
- **Density:** More points were used where the outline changes direction quickly (head, forelegs, mane, tail) and fewer on long gentle runs (back, belly).

4.3 Sharp corners using tripled control points

A silhouette of the Ferrari logo needs genuinely pointed features: hooves, mane tips, ear tips, the elbow, the crotch and the tail tips. With one control point each, the cubic B-spline would round all of them.

To make a point sharp, it was entered three times in a row in the control list. For a uniform cubic B-spline, the curve value at a knot t_j is

$$C(t_j) \;=\; \frac{P_{j-1} + 4\,P_j + P_{j+1}}{6}.$$

If $P_{j-1} = P_j = P_{j+1} = P$, this equals P exactly, so the curve passes through that point. The tangent at the same knot is $(P_{j+1} - P_{j-1})/2$, which is zero. The curve therefore arrives at P and leaves it in a new direction, which appears as a sharp geometric corner.

Between two neighbouring tripled points (for example the front and back of a hoof) the control polygon is a straight line, so the curve there is a straight segment, which gives flat hoof bottoms.

This differs from the method used in the notebook's `RUN_SHARP_DEMO`, which keeps one control point per corner and instead repeats an interior knot $k - 1 = 3$ times to lower continuity at that knot. Both methods produce a visible corner. The tripled-point method was chosen for the horse because it needs no hand-built knot vector: with 206 control points the knot vector would otherwise be long and error-prone to edit, and the open-uniform vector from `generate_open_uniform_knots` can be reused as-is.

Points that are meant to stay softly rounded, such as the fetlock and coronet (where each hoof meets the leg) and the knee bends, were entered once.

4.4 Summary of the final control polygon

Total control points: 206 (122 single points and 28 tripled corner entries, 84 points). The toe of the forward hind hoof at $(72, 0)$ appears as a tripled corner at both the start and the end of the list.

Sharp corner locations (tripled):

- Hooves: forward hind hoof toe $(72, 0)$ and heel $(91, 0)$; rear hind hoof toe $(99, 0)$ and heel $(112, 0)$; lower foreleg hoof $(31, 59)$ and $(20, 58)$; upper foreleg hoof $(14, 95)$ and $(4, 93)$.
- Body: elbow $(50, 84)$ and armpit $(48, 99)$; throat $(46, 124)$; crotch $(96, 62)$; tail root $(118, 93)$.
- Ears: tips at $(44, 164)$ and $(53, 162)$.
- Mane: four tips $(72, 150)$, $(78, 140)$, $(84, 132)$, $(90, 122)$ and three valleys between them $(60, 142)$, $(65, 133)$, $(71, 124)$.
- Tail: three tips $(146, 138)$, $(152, 98)$, $(140, 82)$ and two valleys $(132, 106)$, $(132, 92)$.

5 Knot vector

The default open-uniform vector was used for the outline. For $n = 206$ control points and order $k = 4$:

- The knot vector has $n + k = 210$ entries.
- The first four entries are 0 and the last four are 1, so the curve starts at the first control point and ends at the last.
- There are $210 - 8 = 202$ interior knots, equally spaced at $1/203 \approx 0.00493$.
- There are $n - k + 1 = 203$ knot spans, each a cubic polynomial piece.

Because the interior knots are all simple (multiplicity one), the parametric curve is C^2 continuous everywhere; the geometric corners in Section 4.3 come from the repeated control points, not from lowered continuity. The knot vector is printed by the notebook and is also copied by running the cell.

6 Iterative development

The shape was refined in two stages. Each pass was rendered and inspected before the next.

Stage 1 (passes 1–3) produced a right-facing rearing horse with 109 control points. Within pass 3 there were two internal revisions. The first draft had the forelegs overlapping and tangled, and a small kink in the muzzle caused by a zig-zag in the mouth points. The foreleg points were repositioned so the raised, folded leg sits above and clear of the extended leg, with a notch

between them. The mouth points were smoothed into a single lower-lip line.

Stage 2 (passes 4–6) redesigned the horse to face left and added much more detail. The new points were laid out from scratch rather than mirrored, so that the tail, mane and legs could be reshaped instead of simply flipped.

Pass	Description	Result
1	About 55 points tracing a single-legged rearing horse with a basic head, neck, one foreleg and a tail.	The curve was smooth but read as a blob. The foreleg and head merged.
2	About 55 points with a clearer silhouette: distinct head, arched neck, folded foreleg, flowing tail.	Recognisable as a rearing horse, but simple. Used as the baseline before complexity was increased.
3	109 points, right-facing, with two forelegs, two hind legs, three mane tufts, an ear, nose and chin, and a curled tail tip, using tripled corner points.	Read clearly as a rearing horse. This is the starting point for stage 2.
4	First left-facing draft, 173 control points: forelegs reaching left, head at the upper left, a sawtooth mane and a tail sweeping up and right.	The pose was right, but the torso looked thin, the tail was a messy cluster of spikes, and the rear hind leg was a straight column.
5	Second draft, 195 control points: back and rump reshaped, tail rebuilt as three clean locks, hind legs given a stifle and hock, hooves given a flared shape, mane lengthened into four tufts.	Much more horse-like. The head was still small and plain, and the outline alone lacked fine detail.
6	Final version, 206 control points: fetlocks and coronets added to every hoof, a rounder buttock, a defined jaw and muzzle, and two ears.	The outline closes cleanly, corners are sharp where intended, and the head, mane, legs and tail read clearly. This is the version in the notebook.

For speed, the stage 1 preview used SciPy's `BSpline` with the same degree (3) and an identical open-uniform knot vector. During stage 2 the drafts were rendered with the notebook's own `b_spline_curve` function, so what was previewed is exactly what the notebook produces. The final notebook code was then run end to end to confirm that it runs without error, and the saved figure was inspected.

7 Results

Observations:

- The outline is closed: the curve starts and ends at $(72, 0)$.
- Smooth regions (back, neck, belly, thighs, forearms) are C^2 continuous.
- Sharp features (hooves, ear tips, mane tips, elbow, crotch, tail tips) appear as clean corners

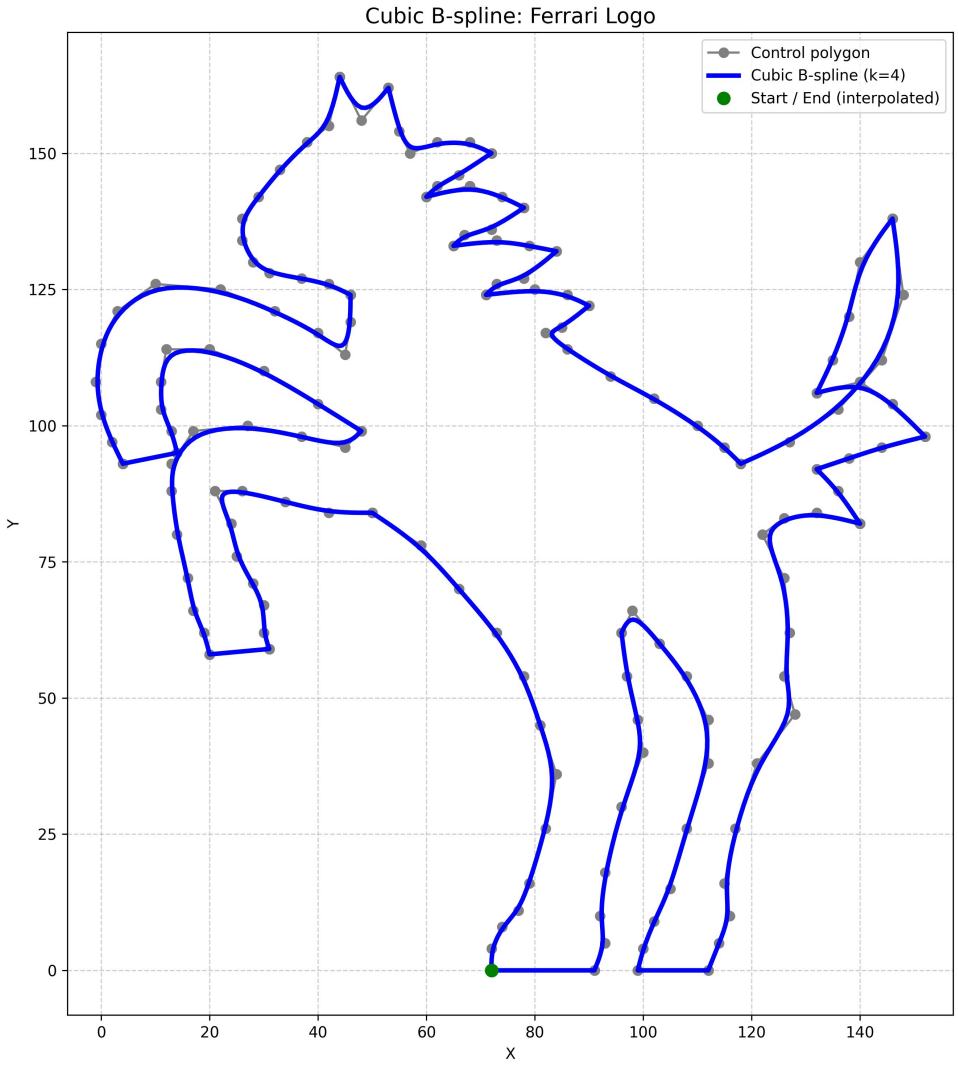


Figure 1: The image `b_spline_curve.jpg` saved by the notebook when the cell is run. It shows the grey control polygon, the blue cubic B-spline outline of the left-facing horse, and the green start/end point at the toe of the forward hind hoof.

where the control points are tripled.

- The flat hoof bottoms come from the straight control-polygon segments between tripled points.

8 Discussion and limitations

- **Approximation, not interpolation.** Placing points on the desired outline does not make the curve pass through them. Overshoot at convex bends was needed, and some features (such as the bend of the rear hind leg and the tail locks) were adjusted by trial and error.
- **Manual tuning.** The control points were chosen by hand. A more automated approach would sample points from a reference image and fit a B-spline (for example with least squares) to the traced outline.
- **Performance.** The provided basis function is a plain recursive Cox–de Boor evaluation.

With 206 control points and 1200 samples it takes a few seconds to run. A de Boor algorithm or vectorised basis evaluation would be faster.

- **Complexity versus control.** More control points allow more detail, but each extra point also gives one more thing that can cause a kink or self-intersection. Local support limits the damage: a bad point only affects four spans.
- **Closure.** The outline closes because the first and last control points coincide. At that hoof toe the curve has a sharp corner anyway, so there is no visible seam. A closed (periodic) B-spline would be needed for a smooth join at an arbitrary point.

9 Conclusion

A 206-control-point cubic B-spline with an open-uniform knot vector produces a closed outline of a left-facing prancing horse, with smooth curves where the body is rounded and sharp corners at the hooves, ear tips, mane tufts, elbow, crotch and tail tips. The sharp corners come from tripling control points, which makes the curve pass through each corner point with zero tangent. The shape was developed in six passes with visual checking at each stage, and it runs in the provided notebook with the B-spline code unchanged.

A Final Control Points

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```

B Final Knot Vector

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```

C Parameters used

Order	$k = 4$ (cubic), degree 3
Outline control points	$n = 206$
Outline knot vector	open-uniform, length 210, interior knot spacing 1/203
Samples along the outline	1200
Code	provided Cox–de Boor implementation, unchanged